

Pei-Chi Pan

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EDUCATION

University of Houston

Ph.D. in Computer Science

Jan 2025 – present

Houston, TX

Research interest: human-centered AI, personal health informatics, human–computer interaction.

Rice University

Master of Computer Science

Dec 2024

Houston, TX

National Yang Ming Chiao Tung University

B.S. in Information Management and Finance

Jun 2023

Hsinchu, Taiwan

Cross-Disciplinary Program in Financial Technology; Minor in English Literature and Linguistics.

PUBLICATIONS

Journal and Conference Papers

[J1] **Pei-Chi Pan**, Yingbin Liang, and Sen Lin. “Reward Modeling for Reinforcement Learning-Based LLM Reasoning: Design, Challenges, and Evaluation.” *Transactions on Machine Learning Research (TMLR)*, 2026.

[C1] Hou, Lu, Karanjai, **Pan**, Lin, Xu, and Shi. “Specialists Hold, Generalists Discount: Asymmetric Equilibrium in LLM Routing Auctions.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.

Under Review

[U1] Hou, Lu, Karanjai, **Pan**, Lin, Xu, and Shi. “Anchors, Mixed Equilibria, and Welfare in LLM Routing Auctions.” Under review, 2026.

RESEARCH EXPERIENCE

Graduate Research Assistant

Department of Computer Science, University of Houston

Jan 2025 – present

Houston, TX

EarMark: Everyday Noise-Sensitivity Sensing with Wearables

- Designing an ecological data-collection protocol that pairs ecological momentary assessment (EMA) with passive Apple Watch sensing to characterize everyday contexts associated with noise-sensitivity reactions.
- Developing the protocol for a participant field deployment (fall 2026) with a privacy-preserving design: no audio or location data is stored.
- Building a multimodal dataset to support personalized models that anticipate difficult moments, extending prior work on assistive technology for noise sensitivity.

Reward Modeling for RL-Based LLM Reasoning [J1]

- Conducted a comprehensive survey of reinforcement learning for LLM reasoning, examining the limitations of chain-of-thought prompting and the shift from RLHF toward verifiable, multi-step reasoning.
- Proposed a taxonomy of reward design to structure the research landscape.
- Analyzed core challenges in RL-for-reasoning pipelines, including reward hacking, diversity collapse, and reasoning hallucinations, and synthesized mitigation strategies.

TEACHING

Teaching Assistant, Machine Learning

Jan 2025 – Dec 2025

Department of Computer Science, University of Houston

Houston, TX

- Led instructional sessions for undergraduates on advanced machine learning topics, including CNNs and LLMs, covering design principles, implementation challenges, and connections to current research.
- Designed and led PyTorch lab sessions; supervised programming projects in which students built and evaluated novel models.

INDUSTRY EXPERIENCE

Microsoft, Azure Cloud & AI

Jul 2022 – Aug 2023

Technical Consultant Intern

Taipei, Taiwan

- Member of a 40-person consulting team that delivered KGI Bank's mobile and internet banking application, Taiwan's first microservices-based internet banking system (launched October 2022).
- Coordinated with developers, product managers, system analysts, and clients on features, testing, and deployment; authored QA test plans and documentation in Azure DevOps; built Angular/TypeScript front-end features.

DBS Bank, Technology & Operations

Jul 2021 – Aug 2021

Data Analyst Intern

Taipei, Taiwan

- Automated customer-transaction retrieval and a daily reporting workflow in Python (pandas) and Microsoft Access, replacing manual processes for the consumer finance team.